Bio Informatics

Lecture 14

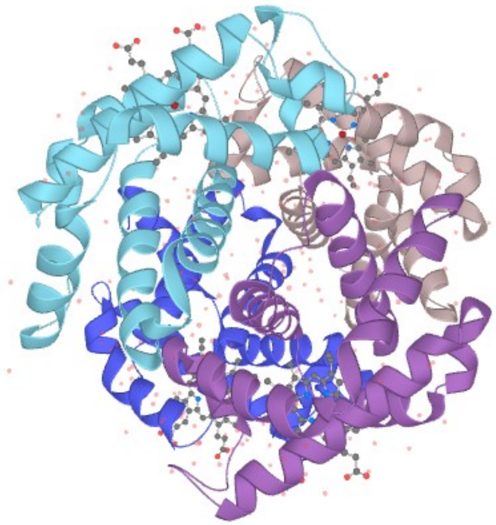
Multiple Sequence Alignments - MSAs

- In Bioinformatics, a sequence alignment is a technique of arranging the sequences of DNA, RNA, or protein to identify regions of similarity.
- For alignment purposes we can align at least two sequences at one time or we can align multiple sequences at one time as well.

Multiple Sequence Alignments - MSAs

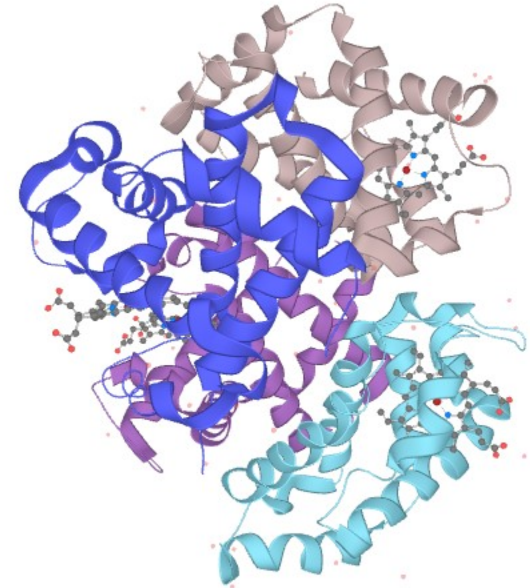
- **Pairwise sequence alignment** methods are used to find the best-matching piecewise (local or global) alignments of **two** query sequences.
- Pairwise alignments can only be used between **two** sequences at a time, but they are efficient to calculate and are often used for methods that do not require extreme precision (such as searching a database for sequences with high similarity to a query).

Hemaglobin subunit alpha - HBA



Human (Homo Sapiens)

MVLSPADKTNVKAAWGKVGAHAGEYGAEALERMFLSFPTT
KTYFPHFDLSHGSAQVKGHGKKVADALTNAVAHVDDMPN
ALSALSDLHAHKLRVDPVNFKLLSHCLLVTLAAHLPAEFTPAV
HASLDKFLASVSTVLTSKYR



Mouse (Mus musculus)

MVLSGEDKSNIKAAWGKIGGHGAEYGAEALERMFASFPTT
KTYFPHFDVSHGSAQVKGHGKKVADALASAAGHLDDLPGA
LSALSDLHAHKLRVDPVNFKLLSHCLLVTLASHHPADFTPAVH
ASLDKFLASVSTVLTSKYR

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HBA_HUMAN          1 MVLSPADKTNVKAAWGKVGAHAGEYGAEALERMFLSFPTTKTYFPHFDLS     50
                     ||||..||:|:||||||:|.|..|||||||||||.|||||||||||||:|
HBA_MOUSE          1 MVLSGEDKSNIKAAWGKIGGHGAEYGAEALERMFASFPTTKTYFPHFDVS     50

HBA_HUMAN         51 HGSAQVKGHGKKVADALTNAVAHVDDMPNALSALSDLHAHKLRVDPVNFK    100
                     |||||||||||||||||.:|..|:||:|.|||||||||||||||||||||
HBA_MOUSE         51 HGSAQVKGHGKKVADALASAAGHLDDLPGALSALSDLHAHKLRVDPVNFK    100

HBA_HUMAN        101 LLSHCLLVTLAAHLPAEFTPAVHASLDKFLASVSTVLTSKYR    142
                     |||||||||||:|.||:|||||||||||||||||||||||||
HBA_MOUSE        101 LLSHCLLVTLASHHPADFTPAVHASLDKFLASVSTVLTSKYR    142
```

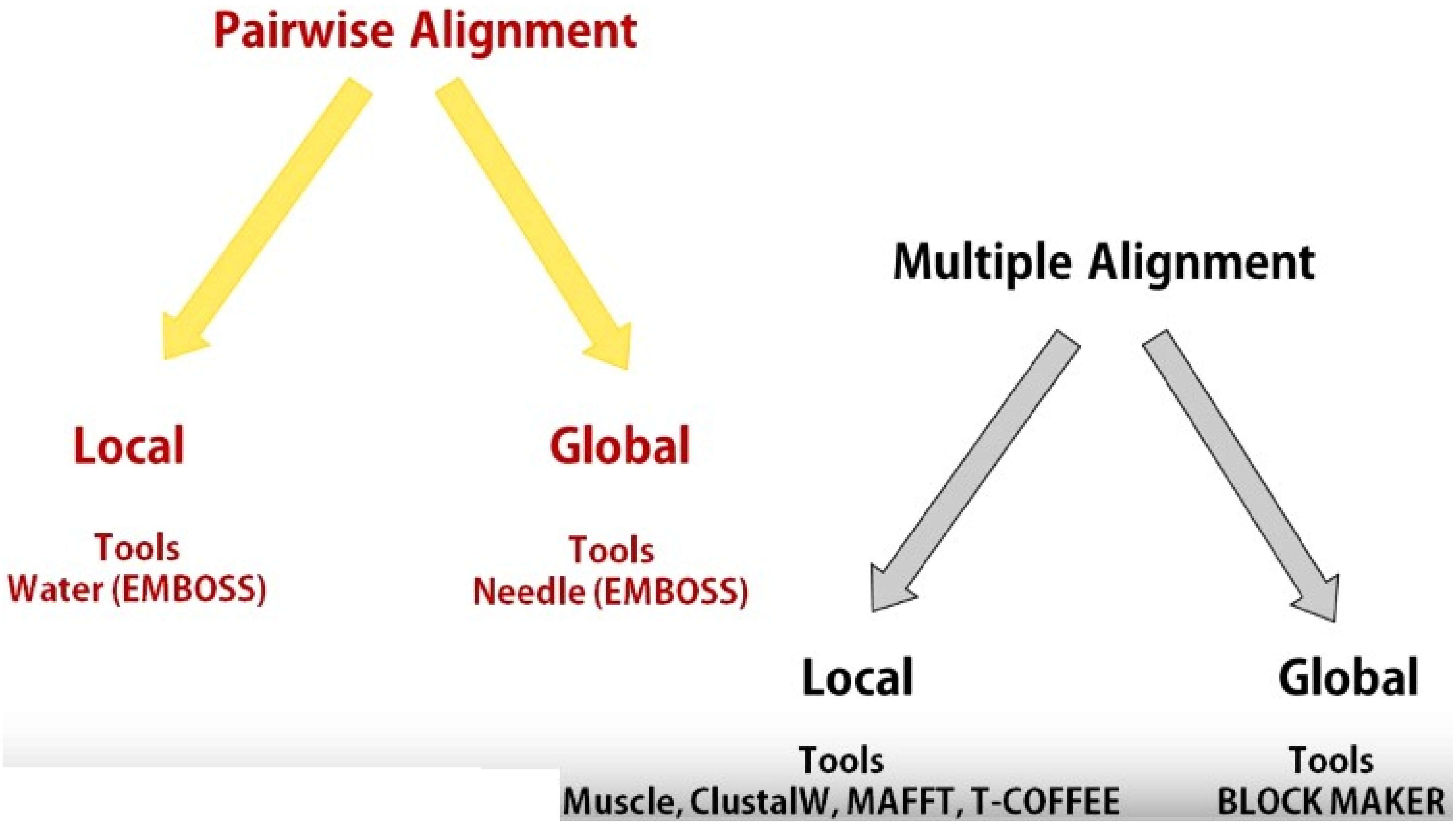
Multiple Sequence Alignments - MSAs

- Across the living organisms, protein performing the same function in different organisms exhibit similar amino acid or gene sequences.
- So in order to find the regions of similarities between them, we perform sequence alignments.
- Through these similarities we can find evolutionary relationships and functional equivalence among two proteins.

Multiple Sequence Alignments - MSAs

- A multiple sequence alignment is basically an alignment of more than two sequences.
- A pairwise alignment tells you about the similarity of two sequences, a multiple alignment tells you about the similarity among multiple sequences.

Sequence Alignment - Types



Multiple Sequence Alignments - MSAs

- **Local** alignments aim to identify small highly similar regions of two proteins or two genes with minimum gaps.
- **Global** alignment aims end-to-end alignment of two sequences, reflecting overall sequence variations.

Different computational approaches to perform a Multiple Alignment

Dynamic programming approach

Basic idea

- This is an extension of dynamic programming approach for pairwise alignment to multiple sequences.
- Unfortunately the cost of computation grows exponentially with the number of sequences (and sequence lengths). This is why this approach is not very effective for multiple alignment.

Progressive alignment method

Step-1

Seq1: AGCCAACCTGGC
Seq2: AATTAATGTA – –

Step-2

Seq1: AGCCAACCTGGC
Seq2: AATTAATGTA – –
Seq3: – – AGGACTTGGC

Step-3

Seq1: AGCCAACCTGGC
Seq2: AATTAATGTA – –
Seq3: – – AGGACTTG – C
Seq4: AACCAACATT – –

Iterative refinement method

Step-1

Seq1: AGCCAACCTGGC
Seq2: AATTAATGTA – –

Step-2

Seq1: AGCCAACCTGGC
Seq2: AATTAATGTA – –
Seq3: – – AGGACTTGGC

Step-3

Seq1: AGCCAACCTGGC
Seq2: AATTAATGTA – –
Seq3: – – AGGACTT$\boxed{\text{G – C}}$
Seq4: AACCAACATT – –

Step-4

Seq1: AGCCAACCTGGC
Seq2: AATTAATGTA – –
Seq3: – – AGGACTT$\boxed{\text{– GC}}$
Seq4: AACCAACATT – –

Refinement step